Divyansh Gupta

me24btech11022@iith.ac.in | divgupta02005@gmail.com | +91 6283868657 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.16
SGGS Collegiate Public School Chandigarh	Class XII (CBSE)	2024	86.40%
Bhavan Vidyalaya Chandigarh	Class X (CBSE)	2022	92.20%

SCHOLASTIC ACHIEVEMENTS

National Science Olympiads (IAPT): Cleared the national Stage-1 qualification cutoff in Chemistry (**Top 5% nationwide**); awarded **Top 10%** Centre Merit Certificates in both **Physics (NSEP)** and **Chemistry (NSEC).**

Ranked 7th (IIT Hyderabad) among 100+ prize-eligible participants in **Optiver Trade-a-thon** -- a multi-IIT quantitative trading simulation with 1000+ participants nationwide

Winner, SIC "Learn in Bharat" 2025 Ideathon -- Pitched a solution in a spontaneously formed, cross-cultural team with Japanese participants

LeetCode - **1574** rating and solved **400+ problems.**

EXPERIENCE

Technical Intern, Sai Enterprises - Volset | May-June 2026

Designed and developed the company's live website **volsetindia.com** using Astro and Tailwind CSS, handling frontend development end-to-end

Modeled CAD designs for stabilizer cabinets using Fusion 360, supporting product design and manufacturing workflows

PROJECTS

Mechanically-Powered Prosthetic Arm*(Ongoing, 5-member team, under Prof.* Safvan Palathingal *)*

Leading a 5-member team designing a full mechanically-powered prosthetic arm for a child, from CAD design through physical assembly and mechanical analysis, under the guidance of Prof. Safvan Palathingal .

- Engineered the current pinching mechanism (cable and torsion-spring driven), benchmarked against commercially available prosthetics. Currently developing static balancing to significantly reduce actuation force and improve usability
- Scoping the mechanism's expansion from the current hand/pinching stage into a complete shoulder-to-hand system with additional degrees of freedom, alongside design refinement for aesthetics
- Hands-on fabrication across 3D printing (slicing/prep), Fusion 360 CAD, and mechanical assembly (drilling, cutting, cable/spring integration)

Advanced Multi-Asset Rotation Engine (LSTMSentimentTrader) | Python, PyTorch, ONNX Runtime, QuantConnect/LEAN, Transformers, Alpaca API, AWS EC2, Docker, cron

- Built a regime-aware, multi-asset trading system combining an LSTM directional predictor with heuristic macro filtering to dynamically rotate leveraged capital (up to 1.5x) across a 6-asset universe (SPY, QQQ, IWM, XLE, GLD, SHV) -- deployed as an autonomous pipeline on AWS EC2, engineered to run ML inference on a memory-constrained micro-instance via swap-based memory management and isolated virtual environments, executing live daily via cron against Alpaca paper trading
- Designed a 12-feature real-time inference pipeline (returns, volatility, RSI, MACD, Bollinger Width, ATR, VIX metrics) feeding an ONNX-optimized LSTM, with momentum-ranking and asset-specific ATR trailing stops for risk management
- Engineered a hybrid sentiment system (automated FinBERT news-scoring pipeline via cron, VIX-proxy fallback) and separately reproduced backtest results in a self-hosted, offline LEAN/Docker environment, achieving parity with cloud backtest results

- Backtested over ~7 years (Sharpe 0.998, 56% win rate, 1,862 orders); validated out-of-sample on the model's genuinely unseen training window (Sharpe 0.431 vs 0.998 in-sample), correctly isolating a regime-specific weakness during rapid crashes rather than overstating robustness

ProductStore -- *Full-Stack Product Management App*

- Designed a RESTful CRUD API using Express and MongoDB, structuring the backend around an MVC pattern to keep the codebase organized as it grew.
- Built the frontend in React, choosing Zustand for state management paired with Chakra UI for the component library.
- Configured the app for single-service deployment, with the Express backend serving the built React frontend.

Thinkboard -- *Full-Stack Notes App*

- Built a notes API with Express and MongoDB, adding Redis-based rate limiting on write endpoints using Upstash.
- Developed the frontend in React with Tailwind CSS and DaisyUI, including toast-based feedback for clear confirmation on user actions.
- Handled environment-specific configuration so the app behaves differently in local development versus production.

POSITIONS OF RESPONSIBILITY

Head of Outreach, Alumni Cell (2026-Present) -- connecting alumni with current students through career talks and mentorship sessions; coordinating alumni engagement for funding initiatives

Design Head, Sports Council (2026-Present) -- designing merchandise for sports events and campus tournaments

Core Team Member, Words & Board Games Club, Lit-Soc (2025-2026) -- organized themed game nights and board game tournaments on campus

RELEVANT COURSES

Programming & Math: Data Structures and Applications, Intro to Programming, Probability & Statistics, Calculus, Differential Equations, Complex Numbers

Core Mechanical: Thermodynamics, Kinematics & Dynamics of Machinery, Heat & Mass Transfer, Manufacturing, Solid Mechanics, Fluid Mechanics

Other: Finance for Engineers, Competitive Strategies for Startups, Introduction to Entrepreneurship, Responsible Leadership

SKILLS

Languages: Python, C, C++(Data Structures and Algorithm) , SQL(DBMS),JavaScript(Object oriented programming)

Tools/Design: MS Office Suite, Matlab, Abaqus , Canva, Figma, Git/GitHub, Vercel, QuantConnect

Web Dev: MERN Stack (React, Node.js, Express, MongoDB), Astro, Tailwind CSS

ML/Data: PyTorch, XGBoost, Scikit-learn, Pandas, NumPy, ONNX

CAD: Fusion 360, SolidWorks

EXTRACURRICULAR

Table Tennis, Inter-IIT Sports Meet -- Represented IIT Hyderabad for 2 consecutive years (2024-2026); reached pre-quarterfinals both editions

Enjoys sketching character art and playing music (flute, keyboard) in free time

Chess -- Casual player on Chess.com, rated 1700+